

SMART DOOR LOCK

Abstract

This project presents the design and implementation of a Smart Door Lock system using a keypad-based password authentication method. The system uses a microcontroller with a keypad for password entry, a servo motor for latch movement, a solenoid door lock for secure locking, and a relay module for switching control. When the correct password is entered, the controller activates the servo motor and solenoid lock to open the door. This system improves security, convenience, and low-cost automation for homes and offices.

Keywords

Smart Door Lock, Keypad, Servo Motor, Solenoid Lock, Relay Module, Arduino, Security System

1. Introduction

Traditional locks require physical keys that can be lost or duplicated. Electronic door locks provide better safety and ease of use. This project develops a password-based smart lock system using a keypad. Authorized users enter a password to unlock the door. Incorrect attempts are denied.

2. System Architecture

Components:

- Arduino / Microcontroller
- 4x4 Keypad
- Servo Motor
- Solenoid Door Lock
- Relay Module
- Power Supply

Working Principle:

1. User enters password on keypad.
2. Controller checks password.
3. If correct, relay activates solenoid lock.
4. Servo motor rotates to open latch.
5. After delay, door locks again.

3. Hardware Components

- Arduino Uno / Nano
- 4x4 Matrix Keypad

- Servo Motor (SG90 / MG995)
- 12V Solenoid Door Lock
- 1 Channel Relay Module
- Adapter / Power Supply
- Jumper Wires
- Buzzer
- LCD

4. Software Requirements

- Arduino IDE
- Embedded C / Arduino Programming

5. Features

- Password protected access
- Automatic locking after delay
- Strong locking using solenoid
- Easy password entry using keypad
- Low cost and reliable

6. Advantages

- No physical key required
- Better security
- Easy to operate
- Suitable for home and office use

7. Applications

- Homes
- Offices
- Labs
- Cabinets
- Hostel rooms

8. Conclusion

The smart door lock using keypad, servo motor, solenoid lock, and relay module is an effective security solution. It is affordable, easy to use, and provides better protection than traditional locks.

9. Literature Review

Many electronic locking systems have been developed using password authentication, RFID cards, fingerprint sensors, and mobile applications. Among these, keypad-based systems remain popular because they are simple, affordable, and reliable. Researchers have found that

combining keypad input with electromechanical locks such as servo motors and solenoids increases both convenience and security. This project uses commonly available components to create a practical locking solution suitable for daily use.

10. Methodology

The system starts in locked mode. The user enters a password through the keypad. Each key press is read by the controller and compared with the stored password. If the entered password matches, the relay module energizes the solenoid lock and the servo motor rotates to release the latch. After a preset time delay, the servo returns to its original position and the relay deactivates the solenoid, locking the door again. If a wrong password is entered, access is denied and buzzer beep 2 times.

11. Circuit Description

The keypad rows and columns are connected to digital pins of the Arduino. The servo motor signal wire is connected to a PWM pin. The relay module input pin is connected to another digital output pin. The relay switches external power to the solenoid lock because the solenoid requires higher current than the Arduino can supply directly. A common ground connection is maintained between all modules for proper operation.

12. Working Flow

1. Power supply is turned on.
2. Controller initializes keypad and output devices.
3. User enters password.
4. Password is verified.
5. If correct, unlock sequence starts.
6. Door remains open for a few seconds.
7. System returns to lock mode.
8. If incorrect, system remains locked and buzzer beep 2 times.

13. Innovation

This project combines two locking actions: a servo motor for latch movement and a solenoid lock for strong holding force. The dual-lock concept improves reliability compared to systems using only one mechanism. It is also low cost and easy to maintain.

14. Limitations

- Password can be guessed if too simple.
- Requires continuous power supply.
- Mechanical parts may wear over time.
- Basic version does not provide remote monitoring.

15. Future Scope

- Add fingerprint authentication.
- Add GSM/Wi-Fi alerts.
- Battery backup during power failure.

16. Testing and Results

The prototype was tested using multiple password entries. Correct passwords successfully activated the relay, solenoid lock, and servo motor. Wrong passwords did not unlock the system. The locking and unlocking response time was quick and stable. The project demonstrated reliable operation for repeated use.

17. Cost Estimation

- Arduino Board
- Keypad Module
- Servo Motor
- Solenoid Lock
- Relay Module
- Adapter and Wires

The total project cost is economical compared to commercial smart locks.

18. References

1. Arduino Official Documentation.
2. Keypad Library Documentation.
3. Embedded Security System Research Papers.
4. Basic Electronics and Relay Control Manuals.

SMART DOOR LOCK USING KEYPAD, SERVO MOTOR, SOLENOID DOOR LOCK AND RELAY MODULE

CIRCUIT DIAGRAM

